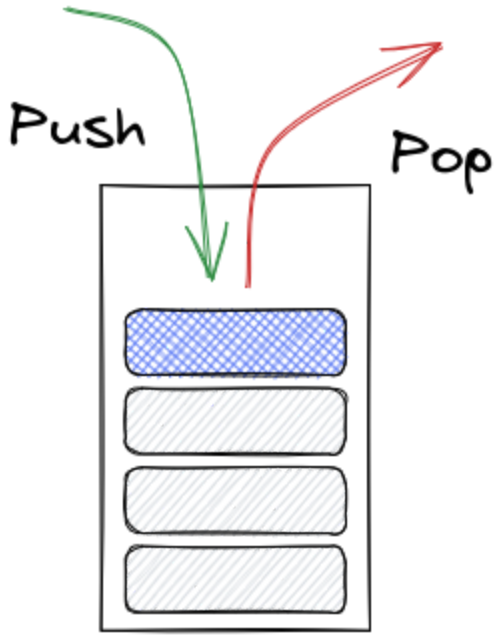
StackLang

LIMITS `2s, 512 MB`

StackLang is a stack-based programming language. Here, each program contains a series of commands that perform various stack operations.

A stack is a last-in-first-out data structure. You can push and pop elements to and from a stack. The element that has been pushed most recently is the top element. The element that comes out when a stack is popped is the element from the top.



The stack used in StackLang can store 32-bit signed integers only and can store at most 10^6 integers.

The language provides various commands to manipulate the stack:

- `PUSH {V}` : Push `{V}` into the stack. When the stack is full (i.e. it has 10^6 elements), the bottom-most element is dropped.
- `POP` : Pop the top element from the stack. No change when the stack is empty.

- `PRINT` : Print the top element from the stack. Prints "-" when the stack is empty.
- `SIZE` : Print the number of elements in the stack. Prints "0" when the stack is empty.
- `SUM` : Print the summation of all the elements in the stack. Prints "0" when the stack is empty.
- `REPEAT {N}` : Repeats all the elements in the stack `{N}` times. If the stack currently has `[1, 3, 7]`, 7 being at the top, then after executing the `REPEAT 1` command, the stack will contain `[1, 3, 7, 1, 3, 7]`. When the stack is full, the bottom-most elements are dropped to make space for new elements from the repeat operation. Logically, a `REPEAT N` operation is like $N \times M$ `PUSH` commands, where M is the number of elements in the stack immediately before the repeat operation begins.
- `REVERSE` : Reverses the entire stack. If the stack currently has `[1, 3, 7]`, 7 being at the top, then after `REVERSE`, the stack will contain `[7, 3, 1]`, 1 being at the top.

Here is an example StackLang program:

```
PUSH 1
PUSH 3
PUSH 7
PRINT
REVERSE
PRINT
```

This example StackLang program will output the following:

```
7
1
```

As you can see, in a well-formatted StackLang program, each command is in a separate line. There are no empty lines. For commands that accept a parameter, there is exactly one space between the command name and the parameter. The command names are always in uppercase.

In this problem, you have to implement an interpreter for StackLang. Easy, right?

INPUT

All StackLang programs in the test cases will have at most 10^6 lines of commands. Each command will be in a separate line. There will be no empty lines. You will have to read until EOF.

Your program should interpret the StackLang program provided in the input and perform actions against each command.

Commands `PUSH` and `REPEAT` take a single argument. There will be exactly one space between the command and the argument. All other commands do not take any argument.

In case of `PUSH {V}`, `{V}` will be a valid 32-bit signed integer.

In case of `REPEAT {N}`, `{N}` will be in the range $[1, 10^6]$ (between 1 and 1000000, inclusive).

OUTPUT

You will print the output of `PRINT`, `SIZE`, and `SUM` commands, each in a separate line, as described in the problem statement.

EXAMPLES

INPUT	OUTPUT
PUSH 1 PUSH 3 PUSH 7 PRINT REVERSE PRINT SIZE POP SUM	7 1 3 10

After the first three operations you have `[1, 3, 7]` in the stack.

The command `PRINT` will then print 7 to the output, 7 being the element at the top of the stack. `REVERSE` will reverse the contents of the stack: `[1, 3, 7]` will become `[7, 3, 1]`. The next `PRINT` command will then print 1 to the output.

`SIZE` will print 3 to the output. `POP` will remove 1 from the stack. `SUM` will print the summation of 7 and 3: 10.

INPUT	OUTPUT
PUSH 1 PUSH 3 PUSH 7 REPEAT 1 PRINT POP PRINT POP PRINT POP PRINT POP PRINT	7 3 1 7 3

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The Assignment

LIMITS `1s, 512 MB`

Bolt is working at a prominent company called Icefire. He has been given a task for evaluation. The task is about substring triplets. A string s of length n is given at first. To build a substring triplet $(s_1,\ s_2,\ s_3)$, the following procedures are followed (1-based indexing):

1. At first, three unique integers $b_1,\ b_2,\ b_3\ (1 \leq b_1,\ b_2,\ b_3 \leq n)$ are selected
2. Then an integer $p\ (1 \leq p \leq n;\ \max(b_1,\ b_2) + p \leq n + 1)$ is chosen such that the following two conditions are satisfied:
   - For each $i\ (0 \leq i < p)$, $s_{b_1+i} = s_{b_2+i}$
   - If $\max(b_1,\ b_2) + p \leq n$, $s_{b_1+p} \neq s_{b_2+p}$

   Let, e_1 be $\min(n,\ b_1 + p)$. The substring s_1 is the substring $s_{b_1}, s_{b_1+1}, \ldots, s_{e_1}$
3. Similarly, another integer $q\ (1 \leq q \leq n;\ \max(b_3, b_2) + q \leq n + 1)$ is chosen such that the following two conditions are satisfied:
   - For each $i\ (0 \leq i < q)$, $s_{b_3+i} = s_{b_2+i}$
   - If $\max(b_3, b_2) + q \leq n$, $s_{b_3+q} \neq s_{b_2+q}$

   Let, e_3 be $\min(n, b_3 + q)$. The substring s_3 is the substring $s_{b_3}, s_{b_3+1}, \ldots, s_{e_3}$
4. Finally, the substring s_2 is generated. Let, e_2 be $\min(n, b_2 + \max(p, q))$. The substring s_2 is the substring $s_{b_2}, s_{b_2+1}, \ldots, s_{e_2}$

The score of the substring triplet is $p \times q$.

For example, let a string s of length $n = 10$ is **abababcxya** and $b_1 = 3, b_2 = 1, b_3 = 10$. According to the conditions above, the value of p will be 4 and so, $e_1 = \min(n, b_1 + p) = \min(10, 3 + 4) = 7$. So, the substring $s_1 =$ **ababc** from **ab<u>ababc</u>xya**. Similarly, the value of q will be 1 and so, $e_3 = \min(n, b_3 + q) = \min(10, 10 + 1) = 10$. So, the substring $s_3 =$ **a**

from **abababcxya**. Finally, the value of e_2 will be $\min(n, b_2 + \max(p, q)) = \min(10, 1 + \max(4, 1)) = 5$. So, the substring $s_2 =$ **ababa** from **ababa**bcxya. The score of this substring triplet $(\mathbf{ababc}, \mathbf{ababa}, \mathbf{a})$ is $p \times q = 4 \times 1 = 4$.

Icefire tells a substring triplet good if the substrings s_1, s_2, s_3 are non-empty and lexicographically $s_1 < s_2 < s_3$ satisfies. Bolt has already generated all possible good substring triplets. Now, he needs to calculate the sum of their scores. Can you calculate it for him?

A substring of a string s is a string that consists of some consecutive characters from string s. A string a is lexicographically smaller than a string b if either a is a prefix of b and $a \neq b$, or in the first position where a and b differ, the string a has a letter that appears earlier in the alphabet than the corresponding letter in b.

INPUT

The first line contains an integer T $(1 \leq T \leq 10^5)$ — the number of test cases.

The second line contains an integer n $(3 \leq n \leq 5 \times 10^5)$ — the length of the string.

The third line contains the string s — which consists of only lowercase English letters.

The sum of n over all test cases will not exceed 5×10^5.

OUTPUT

In each test case, print a single integer in a line, the sum of the scores of the good substring triplets. As the answer can be very large, print it modulo 998244353.

EXAMPLE

INPUT	OUTPUT
2 10 abababcxya	19 2

5 zzzax	

In the second case, the only good substring triplet is: $(\mathbf{za}, \mathbf{zza}, \mathbf{zzz})$. Here, the value of p and q are 1 and 2 respectively. So, the answer is = $(p \times q) \mod 998244353$ = $(1 \times 2) \mod 998244353$ = 2.

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Holy Tree

LIMITS `1s, 512 MB`

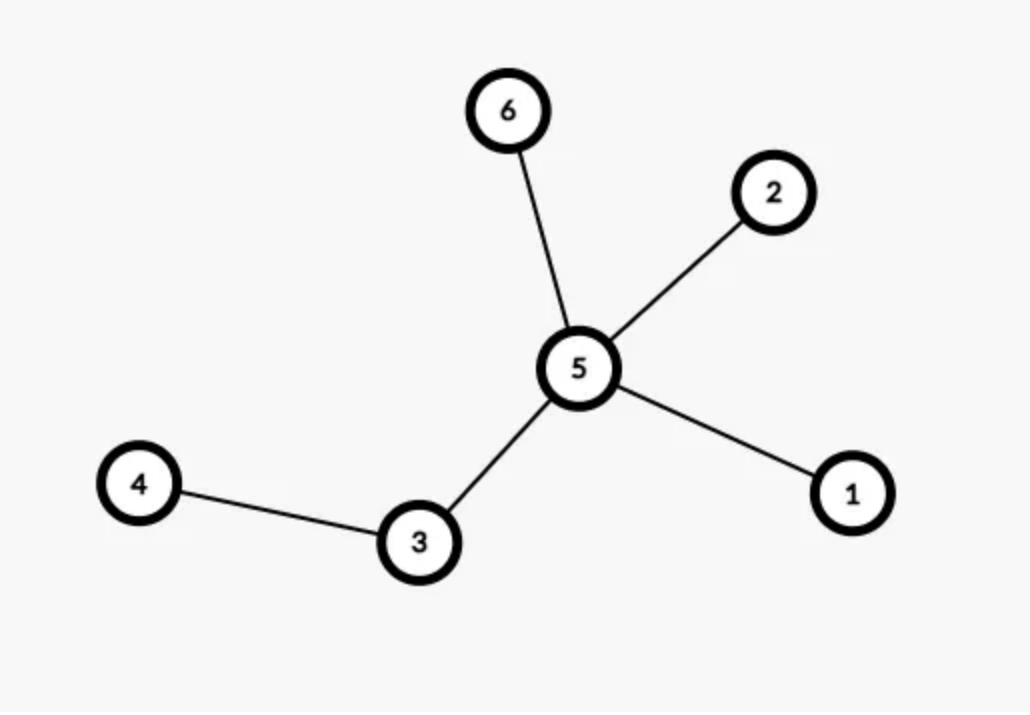
What is a tree?

> A tree is a bidirectional connected graph with no cycle.

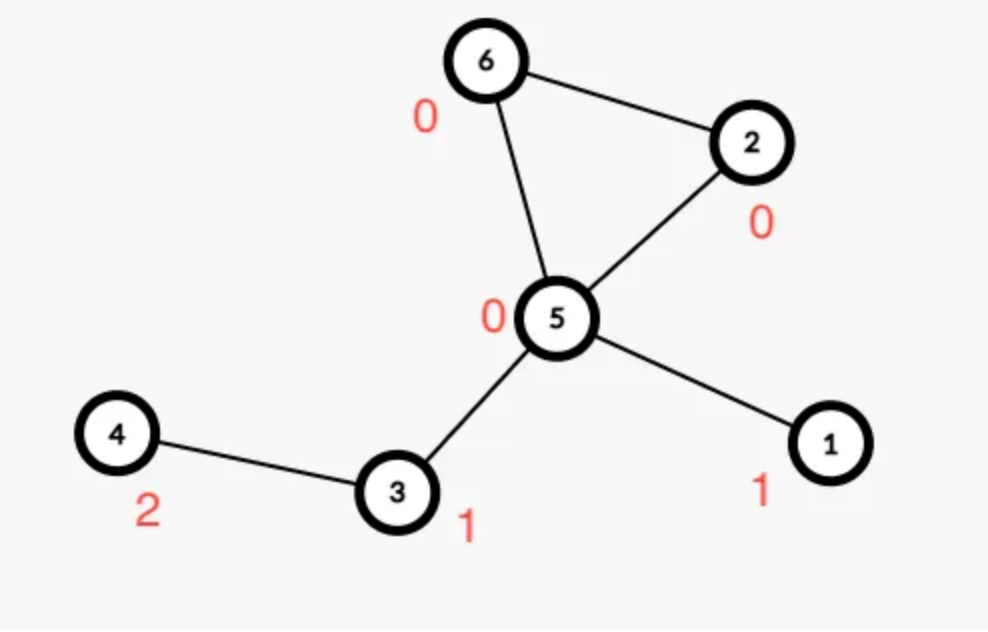
In a tree, we can create exactly one cycle if we connect any two nodes by an edge. Professor Ratman thinks that the cycle has holy powers. So the cost of such one cyclic graph is the total distance from all the nodes to any node of the cycle. Professor Ratman wants to determine the minimum cost given that you can connect any two nodes of the tree.

So you have to minimize $\sum(\text{distance from every node to the cycle})$

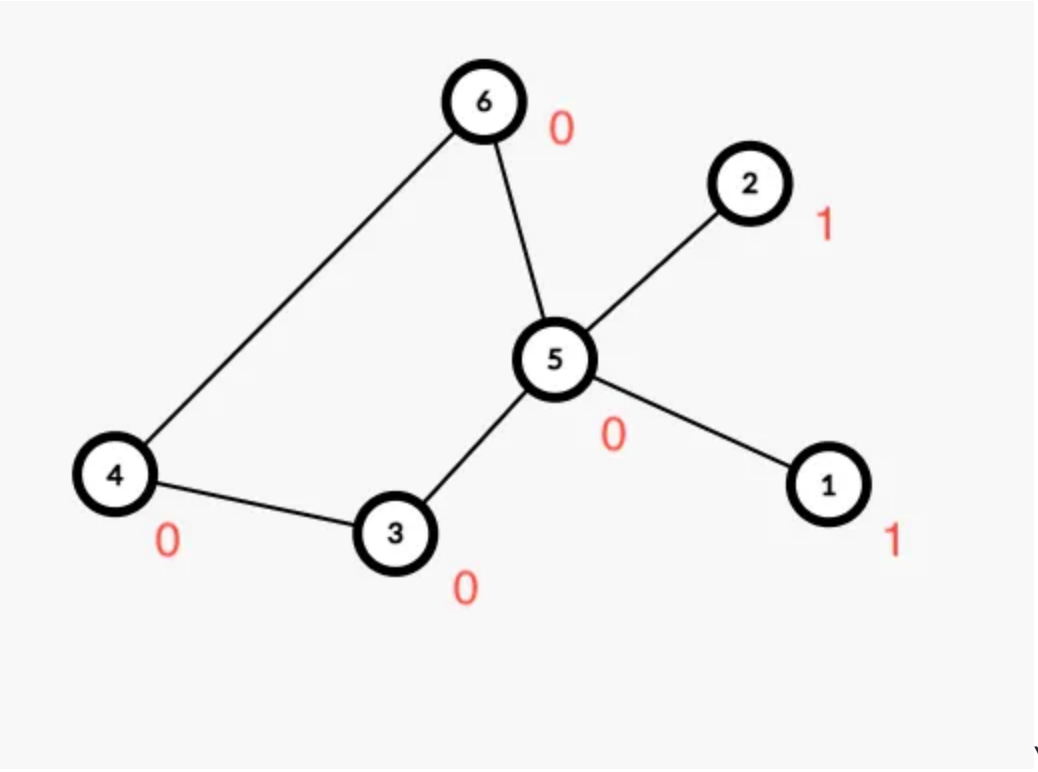
Suppose, we have the following tree:



If we connect 2 and 6 and make a cycle, then the cost would be 1 + 1 + 2 = 4.



But, if we connect 4 and 6, the cost will be, 1 + 1 = 2, which is the minimum cost in this case.



Your job is to find the minimum cost of creating one cycle.

INPUT

The first line will contain an integer $T(1 \leq T \leq 10)$, the number of test cases.

For each test case, the first line contains $n(1 \leq n \leq 10^5)$, the number of nodes in the tree.

Next $n-1$ lines will describe the tree. Each line will contain two space-separated integers u and $v(1 \leq u, v \leq n)$ indicating that there is a bidirectional edge between u and v.

It is guaranteed that the input is a valid tree and the sum of n over all test cases will not exceed 3×10^5, that means $\sum_{t=1}^{T}(n) \leq 3 \times 10^5$.

OUTPUT

For each test case, print "Case X: Y" where "X" indicates the case number and "Y" indicates the minimum cost for the given case.

EXAMPLE

INPUT	OUTPUT
1 6 1 5 5 2 5 6 3 5 3 4	Case 1: 2

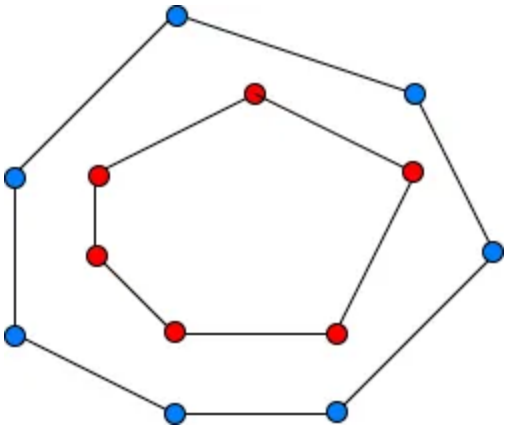
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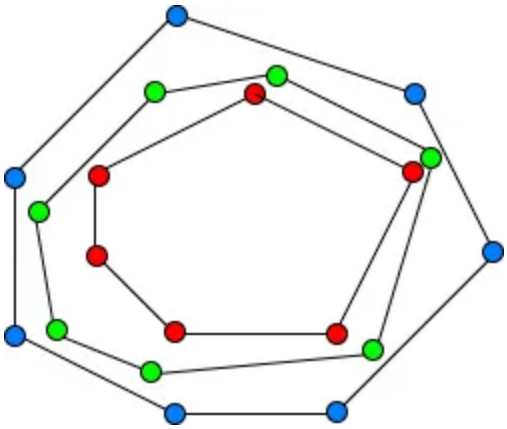
CONSEPHULL

LIMITS `1s, 512 MB`

You are given two convex polygons made out of lattice points (integer coordinates). One of them is completely inside the other one. You have to create another convex polygon that separates the two. The intermediate convex polygon may contain real coordinates but should not contain more than 2 digits after the decimal point.

In other words, build a convex polygon that is completely inside the outer convex polygon and completely outside the inner convex polygon with fulfilling the above constraints. See the figures for a better understanding.





Please, refer to the note section for the definition of a convex polygon and what it means to be strictly inside of a convex polygon.

INPUT

The first line contains two integers n and m where, $3 \leq n, m \leq 30$.

The following n lines each contain two integers x and y representing a point of the outer convex polygon.

Then, the following m lines each contain two integers x and y representing a point of the inner convex polygon.

Each point coordinate range is $0 \leq x, y \leq 10^4$.

Points will be given in a counter-clockwise order for each convex polygon.

OUTPUT

Print an integer p representing the number of points in the new convex polygon. p must be at least 3 and not greater than 1000.

In the following p lines, print the points in counter-clockwise order, forming a valid convex polygon, fulfilling all the requirements in the statement.

Note that, output containing more than 2 digits after the decimal point will receive a **wrong answer** verdict.

EXAMPLE

INPUT	OUTPUT
4 4 0 0 10 0 10 10 0 10 5 1 9 5	4 0.56 0.65 9.35 0.85 9.64 9.00 0.8 9.8

```
5 9
1 5
```

NOTES

- A convex polygon is a simple polygon where all its interior angles are strictly less than 180 degrees.
- A convex polygon is considered completely inside of another convex polygon, when each and every point of the inner polygon is strictly inside of the outer convex polygon. Points on the border of the edges are not considered as strictly inside.

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Diagonal Traversal

LIMITS `1s, 512 MB`

You are given an $n \times m$ grid. You can go from point (x_1, y_1) to any point (x_2, y_2) of grid if and only if x_1 is not equal to x_2 and y_1 is not equal to y_2 and their Euclidian distance is an integer value. You have to tell whether you can go from the top left point $(1, 1)$ to the bottom right point (n, m) using any number of valid moves.

INPUT

The first line of input contains a single integer $T(1 \leq T \leq 1000)$, the number of test cases..

For each test case, there are two integers $n(1 \leq n \leq 10^6)$ and $m(1 \leq m \leq 10^6)$.

OUTPUT

For each test case, print "Case X: S", where "X" is the case number. "S" will be "YES" if it is possible to go from the top left cell $(1,1)$ to the bottom right cell (n,m), and "NO" otherwise.

EXAMPLE

INPUT	OUTPUT
2 1 1 2 2	Case 1: YES Case 2: NO

NOTES

Euclidean distance between two points coordinates (x_1,y_1) and (x_2,y_2) is given by

$$\operatorname{dist}\left(\left(x_1, y_1\right), \left(x_2, y_2\right)\right) = \quad \left(x_1 - x_2\right)^2 + \left(y_1 - y_2\right)^2$$

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Big Tang Theory

LIMITS `1s, 512 MB`

Shell Don (Shelly in short) is a theoretical physicist who is claimed to be highly intelligent with an IQ of 187.

Shelly has decided to drink Tang (a brand of drink mix) regularly. To start out, he made a plan for the next N days. Through rigorous calculations, he decided that the maximum amount of Tang he can drink on the i-th day is A_i. He also decided that he will drink Tang only for a few consecutive days among these N days and on each of those days, he will drink the same amount of Tang.

After hearing this, to make fun of Shelly's intelligence, Pe Nie decided to give Shell Don a math problem. Pe Nie will ask Shelly Q queries. In each query, she will give Shelly a range of days $[L, R]$. Shelly needs to find the maximum total amount of Tang he can drink within some consecutive subset of days from the L-th to R-th day in his schedule (note that he must drink the same amount of Tang on each day of his chosen subset).

More formally, if Shelly's chosen consecutive subset of days starts from the day a and ends at the day b, then $L \leq a \leq b \leq R$ must be satisfied and he must drink the same amount of Tang on each day of the days in $[a, b]$ range. What is the maximum total amount of Tang he can consume?

Despite Shelly's phenomenal intelligence in some sectors, he does have trouble showing true intelligence in many other sectors (especially social common sense and math problems like this). So can you help Shelly in solving this problem so that he is not utterly humiliated in front of Pe Nie?

INPUT

Input starts with two space-separated integers N $(1 \leq N \leq 10^5)$ (the number of days in Shelly's schedule) and Q $(1 \leq Q \leq 10^5)$ (the number of Pe Nie's queries).

In the next line, N space-separated integers are given, the i-th integer A_i $(1 \leq A_i \leq 10^9)$ representing the maximum amount of Tang Shelly can eat in the i-th day.

In each of the following Q lines, two space-separated integers L and R $(1 \leq L \leq R \leq N)$ are given, representing the range of days Pe Nie provides in her queries.

OUTPUT

Print Q lines, where the i-th line contains a single integer representing the answer to Pe Nie's i-th query.

EXAMPLE

INPUT	OUTPUT
5 4 2 5 2 7 9 1 3 1 2 2 4 2 5	6 5 7 14

In the first query, the amount of Tang Shelly can eat on each of the days from the 1st to the 3rd day is 2. So the total amount of Tang he can in that range is 6, this is the maximum possible.

In the second query, if Shelly eats 2 amounts of Tang each day, the total he can eat is 4. But if he eats Tang only on the 2nd day, the total maximum he can eat is 5.

In the third query, Shelly can eat 7 amounts of Tang on the 4th day giving the total maximum amount possible.

In the last query, Shelly can eat 7 amounts of Tang each day from the 4th to the 5th day to obtain a maximum total amount of 14.

Big Tang Theory · toph.co/p/big-tang-theory

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Spreadsheet v1

LIMITS `2.5s, 512 MB`

Are you trying to find a problem that is easy to solve? Even if it involves writing a lot of code? This is the problem you have been looking for!

In this problem, you need to build a tiny spreadsheet application.

You will be given the initial contents of the spreadsheet. For each cell in the spreadsheet, you will be given either a value or a formula. All values are integer numbers. All formulae start with an equal (=) sign, followed by the expression.

Here is what a spreadsheet may look like:

	A	B	C
1	3	15	=A2+B1
2	=B2+A1	2	3
3	6	=A2*A4	6
4	=B3*A3	2	=B4*A4
5	9	8	=B1/B2

All formulae in the spreadsheet will be of the following form: `{CELL1}{OP}{CELL2}`. Here, `{CELL1}` and `{CELL2}` are references to specific cells (can be the same cell), and `{OP}` is an operator (one of `+` addition, `-` subtraction, `*` multiplication, `/` integer division).

Cell references are always of the form `{COL}{ROW}` (e.g. `B2`, `B` is the column alphabet, `2` is the row number). While rows are identified by a number (1-based), columns are identified by one or more alphabets. Column labels go from “A” to “Z”, “AA” to “AZ”, “BA” to “BZ”, “CA”, to “CZ”, …, “ZA” to “ZZ”, “AAA” to “AAZ”, and so on.

You will have to print the entire spreadsheet with the final value of every cell. If the cell contains a value, print it. If the cell contains a formula, evaluate the expression and print the final value.

Final values (after evaluation of expression) should always be one of the following:

- An integer: When the expression can be evaluated.
- "CYCLE": When there is a cycle in the cells referenced in the expression (e.g. cell B3 references A4, which contains a formula that references B3 again).
- "DIVZERO": When there is a division by 0 during the evaluation of an expression.
- “OVERFLOW”: When any operation causes the value of the field to overflow or underflow a 32-bit signed integer. The minimum value a 32-bit signed integer can hold is -2147483648 and the maximum is 2147483647.

When evaluating the formula in any cell, it may happen that through references in that formula you have to jump to different cells, and then eventually arrive back at the cell where you started from. This is a cycle. And all the cells in this cycle should show “CYCLE”.

Any cell that references itself will be result in a “CYCLE”. Any cell that references a cell in a cycle will also show “CYCLE”.

In case the formula in a cell references another cell with either “DIVZERO” or “OVERFLOW”, then the formula will evaluate to an error. In this case the error on the left-hand side of the formula will have higher priority. For example, `=DIVZERO+OVERFLOW` will evaluate to “DIVZERO”. And `=OVERFLOW+DIVZERO` will evaluate to “OVERFLOW”.

If the left-hand side of a formula is an integer or a valid reference, but the right-hand side is an error, the cell will show the same error.

INPUT

The input will start with R ($1 \leq R \leq 1000$) and C ($1 \leq C \leq 1000$), the number of rows and columns in the spreadsheet.

The following R lines each will contain C space-separated tokens. Each token is either a 32-bit signed integer or a string prefixed with "=" (w/o quotes).

Tokens prefixed with "=" will have the format `{CELL1}{OP}{CELL2}`. `{CELL1}` and `{CELL2}` will contain one or more uppercase alphabets followed by an integer. `{OP}` will be one of the following `{+, -, *, /}`.

OUTPUT

Print R lines with C tokens in each (after evaluating all expressions in the given spreadsheet). The tokens must be either integer or one of "CYCLE", "DIVZERO" and "OVERFLOW" (w/o quotes).

EXAMPLES

INPUT	OUTPUT
3 3 1 2 3 4 =A1+C1 5 =A1+A2 =B1+B2 =C1+C2	1 2 3 4 4 5 5 6 8

INPUT	OUTPUT
3 3 1 =A1+C1 1 =A1+B1 =C2+A1 =A2+C3 1 2 =A3+B2	1 2 1 3 CYCLE CYCLE 1 2 CYCLE

INPUT	OUTPUT
3 3 0 1 2 2 =A2/A1 1 2 1 0	0 1 2 2 DIVZERO 1 2 1 0

Spreadsheet v1 · toph.co/p/spreadsheet-v1

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Clock Fantasy

LIMITS `2s, 512 MB`

One day the author imagined an analog clock with minute and hour arms. The thought was simple: if we consider the two arms as two sides of a triangle and join the separate ends of the arms for the third side of the triangle, then at different times throughout the day and night, the area of the triangle varies. A natural question then arises, given the lengths of minute and hour arms, what is the maximum area of the triangle?



In this picture, the blue line segments represent minute and hour arms, and the red line segment represents the third side of the triangle.

INPUT

The first line of the input represents the number of test cases $T(0 < T < 500001)$. The following T lines each contain two floating point numbers a and $b(0 < a, b \leq 10^8)$ representing the length of the hour and the minute arms, respectively.

OUTPUT

For each case, print the maximum area that can be achieved at some position of the hour and minute arms as described above. Errors less than 10^{-6} will be ignored.

EXAMPLE

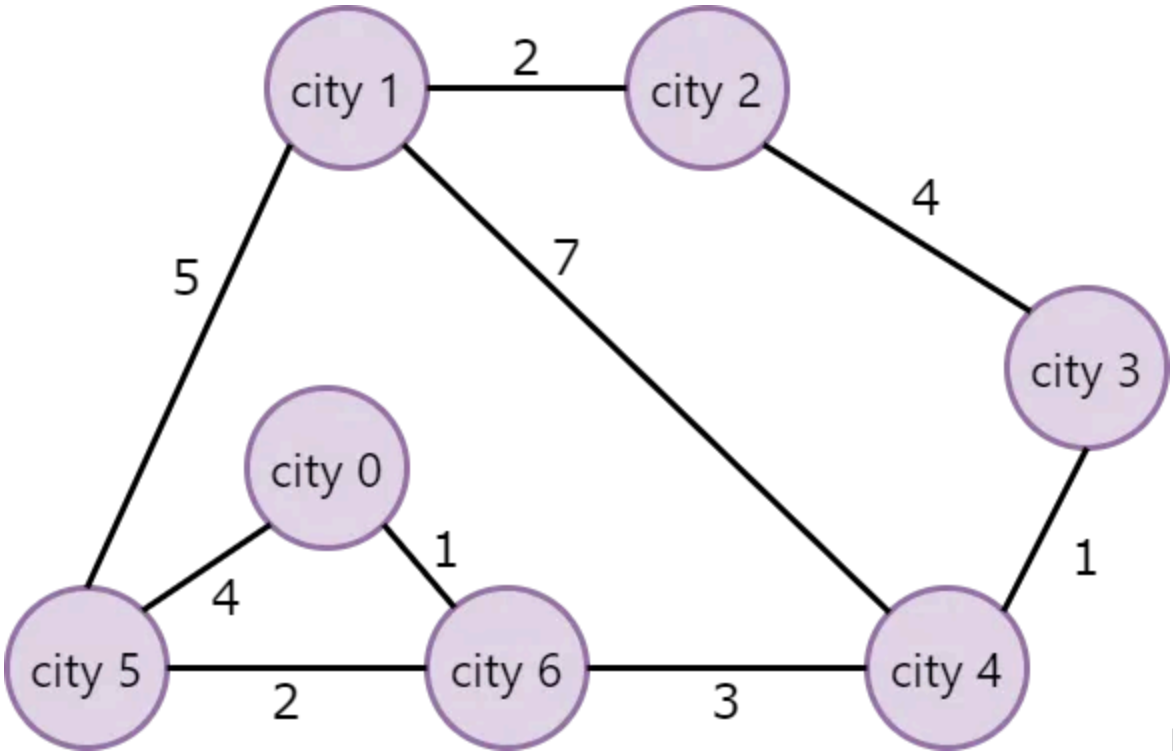
INPUT	OUTPUT
1 661.5611881304 354.5250382063	117270.002748868

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Efficient Construction on Mars

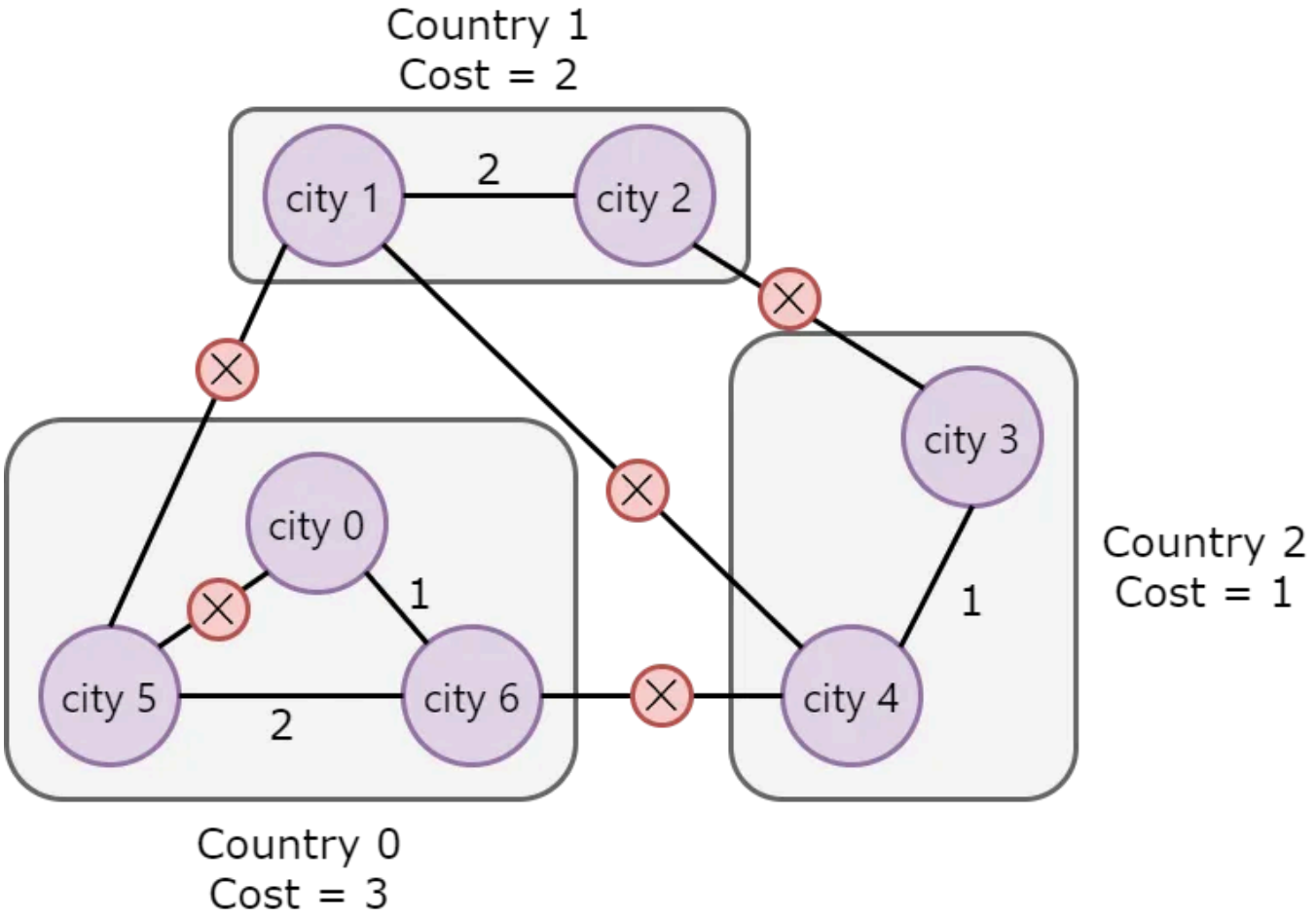
LIMITS `1s, 512 MB`

Elon Musk has constructed N cities numbered from 0 to $N-1$ on Mars. He initially planned to construct E bidirectional roads to connect different cities (where the construction cost of the ith road is W_i. One such example for $N=7$ and $E=9$ can be seen below:



In the figure, the numbers near each road are the cost of constructing that road.

However, Mr. Musk changed his mind a bit. Now, he wants to establish C countries numbered from 0 to $C-1$. Then, he will distribute all the cities among these countries following a special rule. After the distribution, Mr. Musk wants to construct some or all of the E roads he planned initially in such a way that every pair of cities in each country are connected either directly via a road or through another city. Cities from different countries need not be connected. Mr. Musk will distribute the cities so that the cost of constructing such roads is minimized. One such construction of the example shown above can be seen below (for $C=3$):



Note that the roads marked with a $\times$ symbol are not constructed. Here, Mr. Musk constructed 3 countries in such a way that the cost of building roads for each country is minimized. The cumulative costs of building roads for countries 0, 1, and 2 are 3, 2, and 1, respectively.

Now, Mr. Mask has asked you to determine the maximum amount among the cumulative costs of building roads following the rules specified above. In the given example, the maximum cumulative cost is 3.

Given the configuration of the cities and the number of countries, you need to print the maximum cumulative cost. If such construction is not possible, you need to print -1.

It is to be noted that it is possible that a country does not have any city in it.

INPUT

The first line contains three space-separated integers N $(1 \leq N \leq 14)$, C $(1 \leq C \leq 14)$and E $\left(0 \leq E \leq \frac{N \times (N-1)}{2}\right)$ which denote the number of cities, countries, and roads, respectively.

Then E bidirectional roads will be given in the form of three space-separated integers: U_i, V_i $(0 \leq U_i, V_i \leq N-1)$ and W_i $(1 \leq W_i \leq 10^8)$ where U_i and V_i denote the cities at the end of the road and W_i denotes its weight.

OUTPUT

Print the maximum among the minimized cumulative costs. If there is no valid answer, print -1.

EXAMPLE

INPUT	OUTPUT
7 3 9 1 2 2 1 5 5 1 4 7 2 3 4 3 4 1 4 6 3 0 5 4 0 6 1 5 6 2	3

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Order of Digits

LIMITS `1s, 512 MB`

You will be given a list L consisting of N non-negative integers in a decimal number system. We all know that, in the decimal number system, there are 10 digits $(0, 1, 2, 3, 4, 5, 6, 7, 8, 9)$. Their order is defined as $(0 < 1 < 2 < 3 < 4 < 5 < 6 < 7 < 8 < 9)$.

In this problem, you will be given some queries. In each query, you will be given a modified order of the 10 digits as a list P and an integer, X. Here, P denotes the modified order of digits as $P[0] < P[1] < P[2] < P[3] < P[4] < P[5] < P[6] < P[7] < P[8] < P[9]$. The answer to the query is the number of integers that are less than the given integer X in list L (applying the modified digit order described by P). The order of digit 0 will not be modified.

INPUT

First line of the input contains a single integer $N (1 \leq N \leq 10^5)$. The next line will contain N space-separated integers $L_0, L_1, \ldots, L_{N-1}$ which are the elements of the list L and $0 \leq L_i \leq 10^{18}$.

After that, there will be an integer $Q (1 \leq Q \leq 10^5)$ denoting the number of queries. Each of the queries will contain 11 space-separated integers. The first 10 integers will denote the list, P. The 11^{th}integer will denote the value of X $(0 \leq X \leq 10^{18})$. It is guaranteed that $P[0] = 0$ for each of the queries.

OUTPUT

For each test case, print the number of integers that are less than the given integer X in list L applying modified digit priority. Print the answer in a new line.

EXAMPLE

INPUT	OUTPUT
3 10 20 30 1 0 9 8 7 6 5 4 3 2 1 10	2

Given,

List $L = [10, 20, 30]$

Then, there were $Q = 1$ queries are asked.

We can see that the query gives us a list $P = [0, 9, 8, 7, 6, 5, 4, 3, 2, 1]$ and $X = 10$.

From description of the problem, we know the relation given below.

$$P[0] < P[1] < P[2] < P[3] < P[4] < P[5] < P[6] < P[7] < P[8] < P[9]$$

It yields the relation between digits $1, 2$ and 3 as below:

$$3 < 2 < 1$$

So, in the modified order of digits, 30 and 20 are smaller than 10.

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